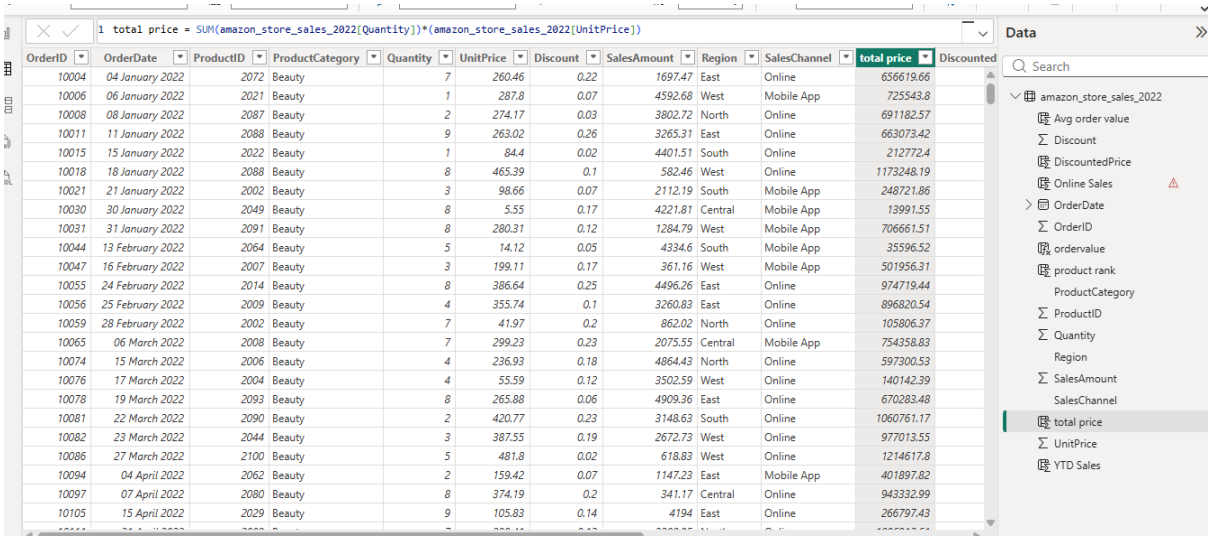
DAX Functions in Power BI Assignment

Tasks to be performed :

1. Create a calculated column totalprice using quantity and unit price.

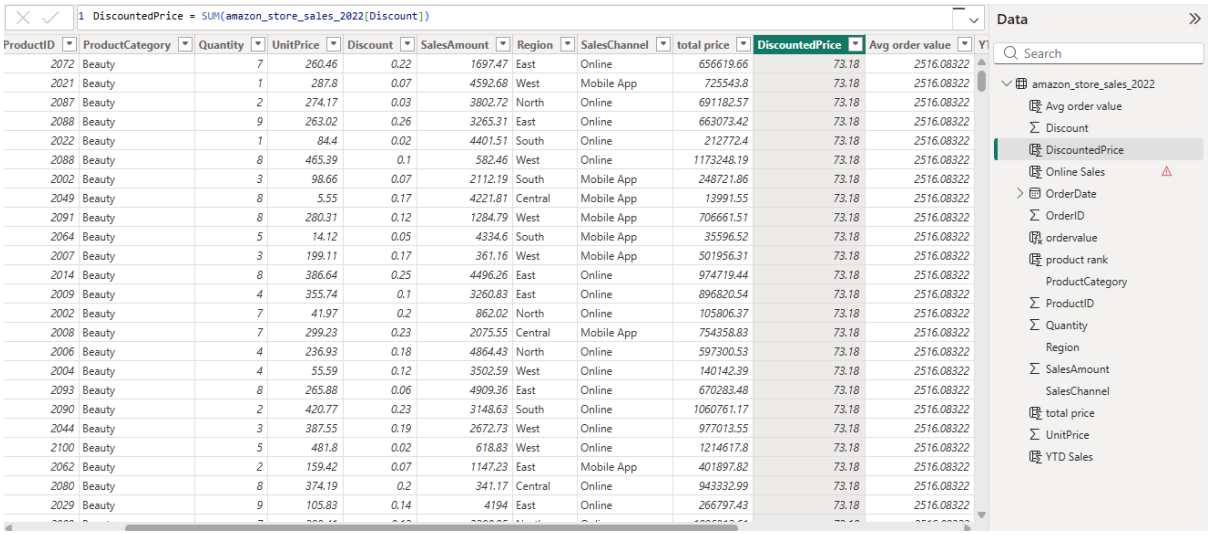
Answer



OrderID	OrderDate	ProductID	ProductCategory	Quantity	UnitPrice	Discount	SalesAmount	Region	SalesChannel	total price	Discounted
10004	04 January 2022	2072	Beauty	7	260.46	0.22	1697.47	East	Online	656619.66	
10006	06 January 2022	2021	Beauty	1	287.8	0.07	4592.68	West	Mobile App	725543.8	
10008	08 January 2022	2087	Beauty	2	274.17	0.03	3802.72	North	Online	691182.57	
10011	11 January 2022	2088	Beauty	9	263.02	0.26	3265.31	East	Online	663073.42	
10015	15 January 2022	2022	Beauty	1	84.4	0.02	4401.51	South	Online	212772.4	
10018	18 January 2022	2088	Beauty	8	465.39	0.1	582.46	West	Online	1173248.19	
10021	21 January 2022	2002	Beauty	3	98.66	0.07	2112.19	South	Mobile App	248721.86	
10030	30 January 2022	2049	Beauty	8	5.55	0.17	4221.81	Central	Mobile App	13991.55	
10031	31 January 2022	2091	Beauty	8	280.31	0.12	1284.79	West	Mobile App	706661.51	
10044	13 February 2022	2064	Beauty	5	14.12	0.05	4334.6	South	Mobile App	35596.52	
10047	16 February 2022	2007	Beauty	3	199.11	0.17	361.16	West	Mobile App	501956.31	
10055	24 February 2022	2014	Beauty	8	386.64	0.25	4496.26	East	Online	974719.44	
10056	25 February 2022	2009	Beauty	4	355.74	0.1	3260.83	East	Online	896820.54	
10059	28 February 2022	2002	Beauty	7	41.97	0.2	862.02	North	Online	105806.37	
10065	06 March 2022	2008	Beauty	7	299.23	0.23	2075.55	Central	Mobile App	754358.83	
10074	15 March 2022	2006	Beauty	4	236.93	0.18	4864.43	North	Online	597300.53	
10076	17 March 2022	2004	Beauty	4	55.59	0.12	3502.59	West	Online	140142.39	
10078	19 March 2022	2093	Beauty	8	265.88	0.06	4909.36	East	Online	670283.48	
10081	22 March 2022	2090	Beauty	2	420.77	0.23	3148.63	South	Online	1060761.17	
10082	23 March 2022	2044	Beauty	3	387.55	0.19	2672.73	West	Online	977013.55	
10086	27 March 2022	2100	Beauty	5	481.8	0.02	618.83	West	Online	1214617.8	
10094	04 April 2022	2062	Beauty	2	159.42	0.07	1147.23	East	Mobile App	401897.82	
10097	07 April 2022	2080	Beauty	8	374.19	0.2	341.17	Central	Online	943332.99	
10105	15 April 2022	2029	Beauty	9	105.83	0.14	4194	East	Online	266797.43	

2.Create a calculated column DiscountedPrice.

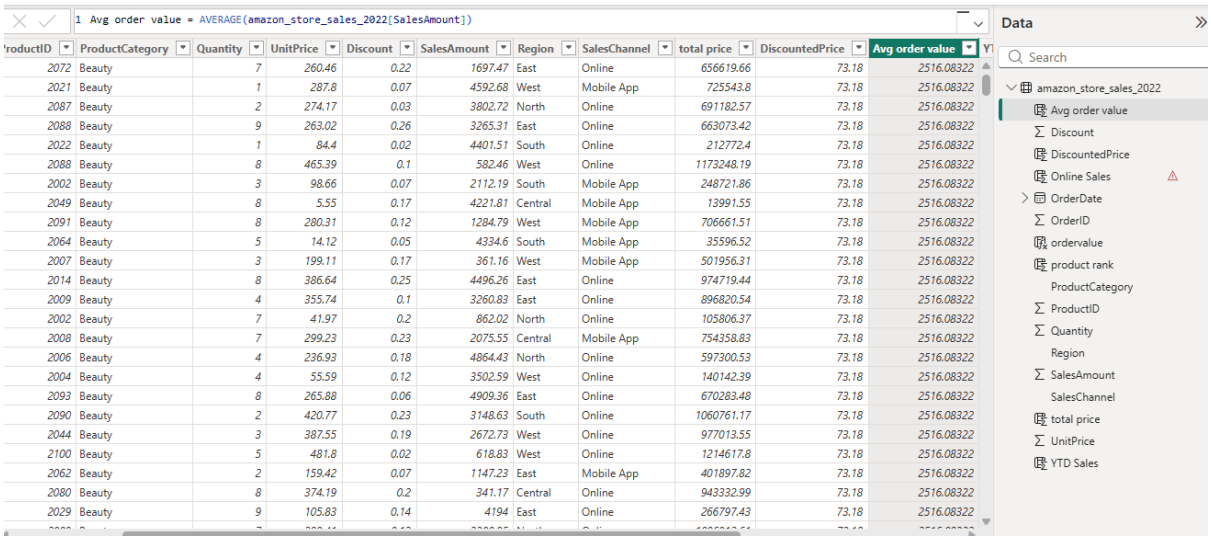
Answer



ProductID	ProductCategory	Quantity	UnitPrice	Discount	SalesAmount	Region	SalesChannel	total price	DiscountedPrice	Avg order value
2072	Beauty	7	260.46	0.22	1697.47	East	Online	656619.66	73.18	2516.08322
2021	Beauty	1	287.8	0.07	4592.68	West	Mobile App	725543.8	73.18	2516.08322
2087	Beauty	2	274.17	0.03	3802.72	North	Online	691182.57	73.18	2516.08322
2088	Beauty	9	263.02	0.26	3265.31	East	Online	663073.42	73.18	2516.08322
2022	Beauty	1	84.4	0.02	4401.51	South	Online	212772.4	73.18	2516.08322
2088	Beauty	8	465.39	0.1	582.46	West	Online	1173248.19	73.18	2516.08322
2002	Beauty	3	98.66	0.07	2112.19	South	Mobile App	248721.86	73.18	2516.08322
2049	Beauty	8	5.55	0.17	4221.81	Central	Mobile App	13991.55	73.18	2516.08322
2091	Beauty	8	280.31	0.12	1284.79	West	Mobile App	706661.51	73.18	2516.08322
2064	Beauty	5	14.12	0.05	4334.6	South	Mobile App	35596.52	73.18	2516.08322
2007	Beauty	3	199.11	0.17	361.16	West	Mobile App	501956.31	73.18	2516.08322
2014	Beauty	8	386.64	0.25	4496.26	East	Online	974719.44	73.18	2516.08322
2009	Beauty	4	355.74	0.1	3260.83	East	Online	896820.54	73.18	2516.08322
2002	Beauty	7	41.97	0.2	862.02	North	Online	105806.37	73.18	2516.08322
2008	Beauty	7	299.23	0.23	2075.55	Central	Mobile App	754358.83	73.18	2516.08322
2006	Beauty	4	236.93	0.18	4864.43	North	Online	597300.53	73.18	2516.08322
2004	Beauty	4	55.59	0.12	3502.59	West	Online	140142.39	73.18	2516.08322
2093	Beauty	8	265.88	0.06	4909.36	East	Online	670283.48	73.18	2516.08322
2090	Beauty	2	420.77	0.23	3148.63	South	Online	1060761.17	73.18	2516.08322
2044	Beauty	3	387.55	0.19	2672.73	West	Online	977013.55	73.18	2516.08322
2100	Beauty	5	481.8	0.02	618.83	West	Online	1214617.8	73.18	2516.08322
2062	Beauty	2	159.42	0.07	1147.23	East	Mobile App	401897.82	73.18	2516.08322
2080	Beauty	8	374.19	0.2	341.17	Central	Online	943332.99	73.18	2516.08322
2029	Beauty	9	105.83	0.14	4194	East	Online	266797.43	73.18	2516.08322

3.Create a measure Average Order Value (AOV).

Answer



```
1 Avg order value = AVERAGE(amazon_store_sales_2022[SalesAmount])
```

roductID	ProductCategory	Quantity	UnitPrice	Discount	SalesAmount	Region	SalesChannel	total price	DiscountedPrice	Avg order value	Y
2072	Beauty	7	260.46	0.22	1697.47	East	Online	656619.66	73.18	2516.08322	
2021	Beauty	1	287.8	0.07	4592.68	West	Mobile App	725543.8	73.18	2516.08322	
2087	Beauty	2	274.17	0.03	3802.72	North	Online	691182.57	73.18	2516.08322	
2088	Beauty	9	263.02	0.26	3265.31	East	Online	663073.42	73.18	2516.08322	
2022	Beauty	1	84.4	0.02	4401.51	South	Online	212772.4	73.18	2516.08322	
2088	Beauty	8	465.39	0.1	582.46	West	Online	1173248.19	73.18	2516.08322	
2002	Beauty	3	98.66	0.07	2112.19	South	Mobile App	248721.86	73.18	2516.08322	
2049	Beauty	8	5.55	0.17	4221.81	Central	Mobile App	13991.55	73.18	2516.08322	
2091	Beauty	8	280.31	0.12	1284.79	West	Mobile App	706661.51	73.18	2516.08322	
2064	Beauty	5	14.12	0.05	4334.6	South	Mobile App	35596.52	73.18	2516.08322	
2007	Beauty	3	199.11	0.17	361.16	West	Mobile App	501956.31	73.18	2516.08322	
2014	Beauty	8	386.64	0.25	4496.26	East	Online	974719.44	73.18	2516.08322	
2009	Beauty	4	355.74	0.1	3260.83	East	Online	896820.54	73.18	2516.08322	
2002	Beauty	7	41.97	0.2	862.02	North	Online	105806.37	73.18	2516.08322	
2008	Beauty	7	299.23	0.23	2075.55	Central	Mobile App	754358.83	73.18	2516.08322	
2006	Beauty	4	236.93	0.18	4864.43	North	Online	597300.53	73.18	2516.08322	
2004	Beauty	4	55.59	0.12	3502.59	West	Online	140142.39	73.18	2516.08322	
2093	Beauty	8	265.88	0.06	4909.36	East	Online	670283.48	73.18	2516.08322	
2090	Beauty	2	420.77	0.23	3148.63	South	Online	1060761.17	73.18	2516.08322	
2044	Beauty	3	387.55	0.19	2672.73	West	Online	977013.55	73.18	2516.08322	
2100	Beauty	5	481.8	0.02	618.83	West	Online	1214617.8	73.18	2516.08322	
2062	Beauty	2	159.42	0.07	1147.23	East	Mobile App	401897.82	73.18	2516.08322	
2080	Beauty	8	374.19	0.2	341.17	Central	Online	943332.99	73.18	2516.08322	
2029	Beauty	9	105.83	0.14	4194	East	Online	266797.43	73.18	2516.08322	

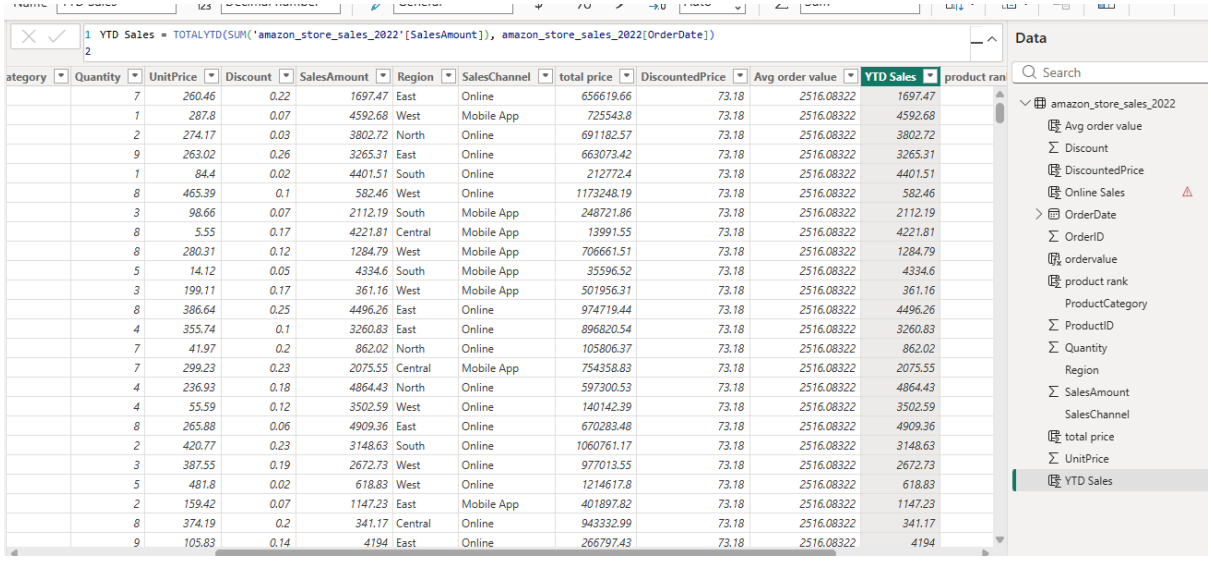
Data

Search

- amazon_store_sales_2022
  - Avg order value
  - Discount
  - DiscountedPrice
  - Online Sales
  - OrderDate
  - OrderID
  - ordervalue
  - product rank
  - ProductCategory
  - ProductID
  - Quantity
  - Region
  - SalesAmount
  - SalesChannel
  - total price
  - UnitPrice
  - YTD Sales

4.Using DAX, create a measure that calculates cumulative Year-to-Date (YTD) Sales based on the Order Date.

Answer



```
1 YTD Sales = TOTALYTD(SUM('amazon_store_sales_2022'[SalesAmount]), amazon_store_sales_2022[OrderDate])
2
```

ategory	Quantity	UnitPrice	Discount	SalesAmount	Region	SalesChannel	total price	DiscountedPrice	Avg order value	YTD Sales	product ran
	7	260.46	0.22	1697.47	East	Online	656619.66	73.18	2516.08322	1697.47	
	1	287.8	0.07	4592.68	West	Mobile App	725543.8	73.18	2516.08322	4592.68	
	2	274.17	0.03	3802.72	North	Online	691182.57	73.18	2516.08322	3802.72	
	9	263.02	0.26	3265.31	East	Online	663073.42	73.18	2516.08322	3265.31	
	1	84.4	0.02	4401.51	South	Online	212772.4	73.18	2516.08322	4401.51	
	8	465.39	0.1	582.46	West	Online	1173248.19	73.18	2516.08322	582.46	
	3	98.66	0.07	2112.19	South	Mobile App	248721.86	73.18	2516.08322	2112.19	
	8	5.55	0.17	4221.81	Central	Mobile App	13991.55	73.18	2516.08322	4221.81	
	8	280.31	0.12	1284.79	West	Mobile App	706661.51	73.18	2516.08322	1284.79	
	5	14.12	0.05	4334.6	South	Mobile App	35596.52	73.18	2516.08322	4334.6	
	3	199.11	0.17	361.16	West	Mobile App	501956.31	73.18	2516.08322	361.16	
	8	386.64	0.25	4496.26	East	Online	974719.44	73.18	2516.08322	4496.26	
	4	355.74	0.1	3260.83	East	Online	896820.54	73.18	2516.08322	3260.83	
	7	41.97	0.2	862.02	North	Online	105806.37	73.18	2516.08322	862.02	
	7	299.23	0.23	2075.55	Central	Mobile App	754358.83	73.18	2516.08322	2075.55	
	4	236.93	0.18	4864.43	North	Online	597300.53	73.18	2516.08322	4864.43	
	4	55.59	0.12	3502.59	West	Online	140142.39	73.18	2516.08322	3502.59	
	8	265.88	0.06	4909.36	East	Online	670283.48	73.18	2516.08322	4909.36	
	2	420.77	0.23	3148.63	South	Online	1060761.17	73.18	2516.08322	3148.63	
	3	387.55	0.19	2672.73	West	Online	977013.55	73.18	2516.08322	2672.73	
	5	481.8	0.02	618.83	West	Online	1214617.8	73.18	2516.08322	618.83	
	2	159.42	0.07	1147.23	East	Mobile App	401897.82	73.18	2516.08322	1147.23	
	8	374.19	0.2	341.17	Central	Online	943332.99	73.18	2516.08322	341.17	
	9	105.83	0.14	4194	East	Online	266797.43	73.18	2516.08322	4194	

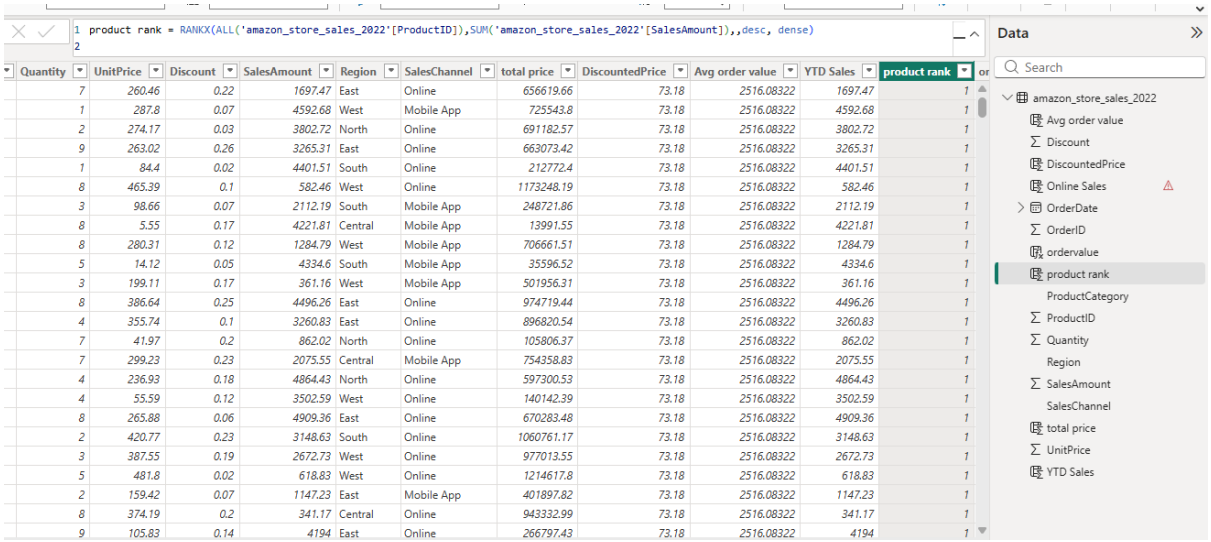
Data

Search

- amazon_store_sales_2022
  - Avg order value
  - Discount
  - DiscountedPrice
  - Online Sales
  - OrderDate
  - OrderID
  - ordervalue
  - product rank
  - ProductCategory
  - ProductID
  - Quantity
  - Region
  - SalesAmount
  - SalesChannel
  - total price
  - UnitPrice
  - YTD Sales

5.Using DAX, create a measure to rank each product according to its Total Sales, with the highest-selling product ranked first.

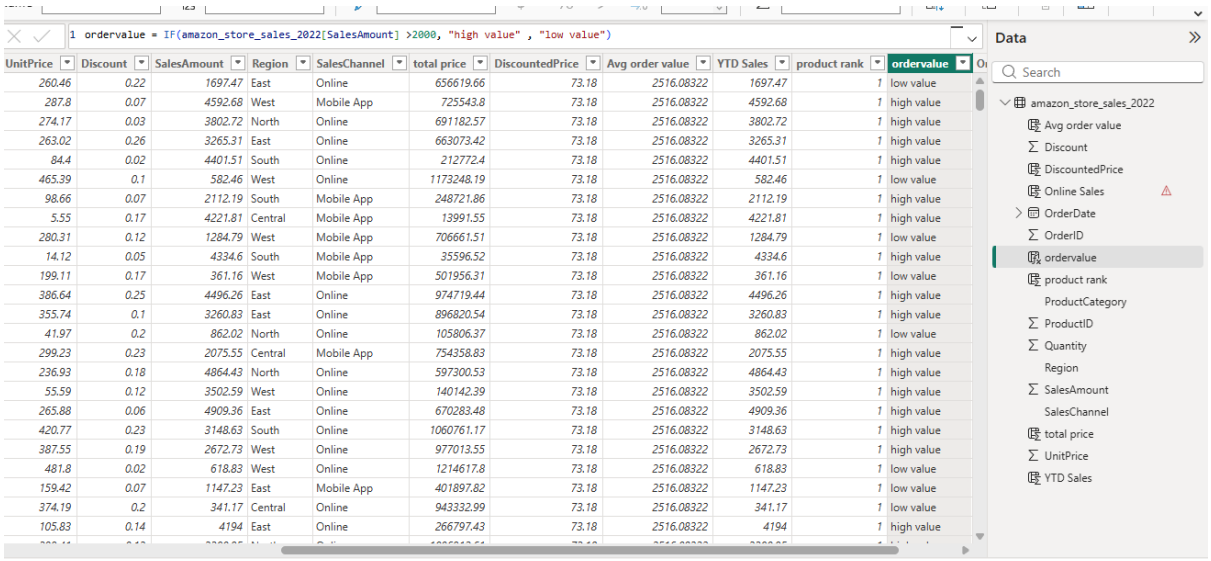
Answer



Quantity	UnitPrice	Discount	SalesAmount	Region	SalesChannel	total price	DiscountedPrice	Avg order value	YTD Sales	product rank
7	260.46	0.22	1697.47	East	Online	656619.66	73.18	2516.08322	1697.47	1
1	287.8	0.07	4592.68	West	Mobile App	725543.8	73.18	2516.08322	4592.68	1
2	274.17	0.03	3802.72	North	Online	691182.57	73.18	2516.08322	3802.72	1
9	263.02	0.26	3265.31	East	Online	663073.42	73.18	2516.08322	3265.31	1
1	84.4	0.02	4401.51	South	Online	212772.4	73.18	2516.08322	4401.51	1
8	465.39	0.1	582.46	West	Online	1173248.19	73.18	2516.08322	582.46	1
3	98.66	0.07	2112.19	South	Mobile App	248721.86	73.18	2516.08322	2112.19	1
8	5.55	0.17	4221.81	Central	Mobile App	13991.55	73.18	2516.08322	4221.81	1
8	280.31	0.12	1284.79	West	Mobile App	706661.51	73.18	2516.08322	1284.79	1
5	14.12	0.05	4334.6	South	Mobile App	35596.52	73.18	2516.08322	4334.6	1
3	199.11	0.17	361.16	West	Mobile App	501956.31	73.18	2516.08322	361.16	1
8	386.64	0.25	4496.26	East	Online	974719.44	73.18	2516.08322	4496.26	1
4	355.74	0.1	3260.83	East	Online	896820.54	73.18	2516.08322	3260.83	1
7	41.97	0.2	862.02	North	Online	105806.37	73.18	2516.08322	862.02	1
7	299.23	0.23	2075.55	Central	Mobile App	754358.83	73.18	2516.08322	2075.55	1
4	236.93	0.18	4864.43	North	Online	597300.53	73.18	2516.08322	4864.43	1
4	55.59	0.12	3502.59	West	Online	140142.39	73.18	2516.08322	3502.59	1
8	265.88	0.06	4909.36	East	Online	670283.48	73.18	2516.08322	4909.36	1
2	420.77	0.23	3148.63	South	Online	1060761.17	73.18	2516.08322	3148.63	1
3	387.55	0.19	2672.73	West	Online	977013.55	73.18	2516.08322	2672.73	1
5	481.8	0.02	618.83	West	Online	1214617.8	73.18	2516.08322	618.83	1
2	159.42	0.07	1147.23	East	Mobile App	401897.82	73.18	2516.08322	1147.23	1
8	374.19	0.2	341.17	Central	Online	943332.99	73.18	2516.08322	341.17	1
9	105.83	0.14	4194	East	Online	266797.43	73.18	2516.08322	4194	1

6.Create a calculated column classifying orders as “High Value” if sales > 2000, else “Low Value”.

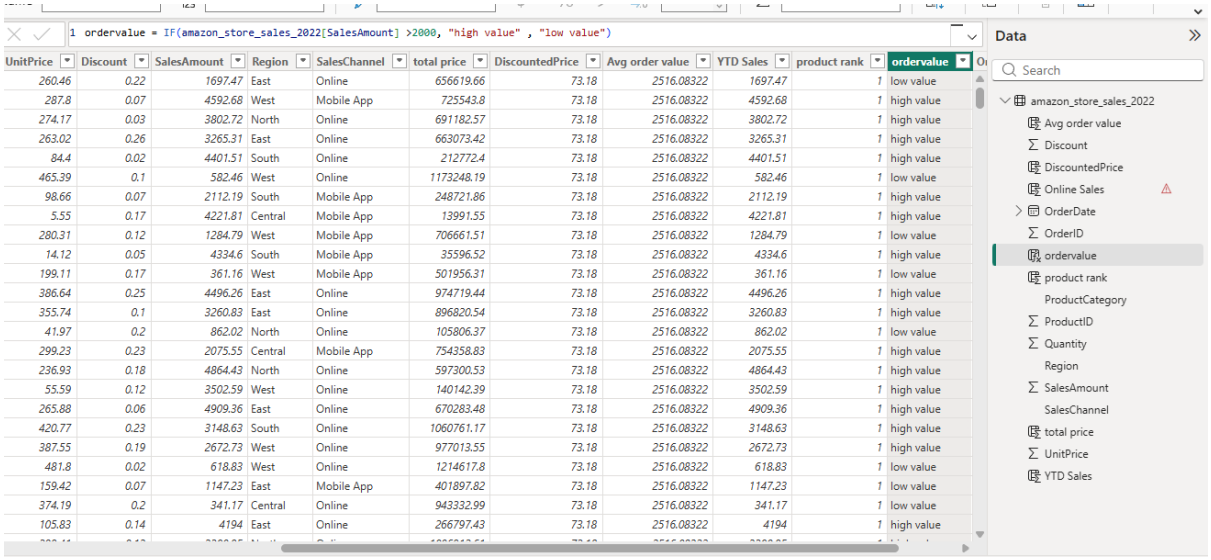
Answer



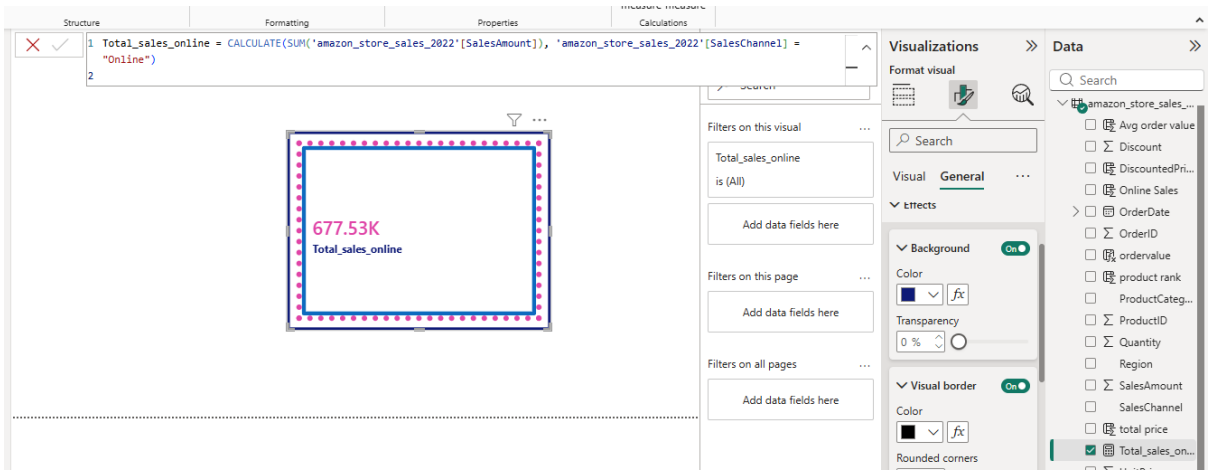
UnitPrice	Discount	SalesAmount	Region	SalesChannel	total price	DiscountedPrice	Avg order value	YTD Sales	product rank	ordervalue
260.46	0.22	1697.47	East	Online	656619.66	73.18	2516.08322	1697.47	1	low value
287.8	0.07	4592.68	West	Mobile App	725543.8	73.18	2516.08322	4592.68	1	high value
274.17	0.03	3802.72	North	Online	691182.57	73.18	2516.08322	3802.72	1	high value
263.02	0.26	3265.31	East	Online	663073.42	73.18	2516.08322	3265.31	1	high value
84.4	0.02	4401.51	South	Online	212772.4	73.18	2516.08322	4401.51	1	high value
465.39	0.1	582.46	West	Online	1173248.19	73.18	2516.08322	582.46	1	low value
98.66	0.07	2112.19	South	Mobile App	248721.86	73.18	2516.08322	2112.19	1	high value
5.55	0.17	4221.81	Central	Mobile App	13991.55	73.18	2516.08322	4221.81	1	high value
280.31	0.12	1284.79	West	Mobile App	706661.51	73.18	2516.08322	1284.79	1	low value
14.12	0.05	4334.6	South	Mobile App	35596.52	73.18	2516.08322	4334.6	1	high value
199.11	0.17	361.16	West	Mobile App	501956.31	73.18	2516.08322	361.16	1	low value
386.64	0.25	4496.26	East	Online	974719.44	73.18	2516.08322	4496.26	1	high value
355.74	0.1	3260.83	East	Online	896820.54	73.18	2516.08322	3260.83	1	high value
41.97	0.2	862.02	North	Online	105806.37	73.18	2516.08322	862.02	1	low value
299.23	0.23	2075.55	Central	Mobile App	754358.83	73.18	2516.08322	2075.55	1	high value
236.93	0.18	4864.43	North	Online	597300.53	73.18	2516.08322	4864.43	1	high value
55.59	0.12	3502.59	West	Online	140142.39	73.18	2516.08322	3502.59	1	high value
265.88	0.06	4909.36	East	Online	670283.48	73.18	2516.08322	4909.36	1	high value
420.77	0.23	3148.63	South	Online	1060761.17	73.18	2516.08322	3148.63	1	high value
387.55	0.19	2672.73	West	Online	977013.55	73.18	2516.08322	2672.73	1	high value
481.8	0.02	618.83	West	Online	1214617.8	73.18	2516.08322	618.83	1	low value
159.42	0.07	1147.23	East	Mobile App	401897.82	73.18	2516.08322	1147.23	1	low value
374.19	0.2	341.17	Central	Online	943332.99	73.18	2516.08322	341.17	1	low value
105.83	0.14	4194	East	Online	266797.43	73.18	2516.08322	4194	1	high value

7.Create a measure to calculate total sales from the Online channel only.

Answer



UnitPrice	Discount	SalesAmount	Region	SalesChannel	total price	DiscountedPrice	Avg order value	YTD Sales	product rank	ordervalue
260.46	0.22	1697.47	East	Online	656619.66	73.18	2516.08322	1697.47	1	low value
287.8	0.07	4592.68	West	Mobile App	725543.8	73.18	2516.08322	4592.68	1	high value
274.17	0.03	3802.72	North	Online	691182.57	73.18	2516.08322	3802.72	1	high value
263.02	0.26	3265.31	East	Online	663073.42	73.18	2516.08322	3265.31	1	high value
84.4	0.02	4401.51	South	Online	212772.4	73.18	2516.08322	4401.51	1	high value
465.39	0.1	582.46	West	Online	1173248.19	73.18	2516.08322	582.46	1	low value
98.66	0.07	2112.19	South	Mobile App	248721.86	73.18	2516.08322	2112.19	1	high value
5.55	0.17	4221.81	Central	Mobile App	13991.55	73.18	2516.08322	4221.81	1	high value
280.31	0.12	1284.79	West	Mobile App	706661.51	73.18	2516.08322	1284.79	1	low value
14.12	0.05	4334.6	South	Mobile App	35596.52	73.18	2516.08322	4334.6	1	high value
199.11	0.17	361.16	West	Mobile App	501956.31	73.18	2516.08322	361.16	1	low value
386.64	0.25	4496.26	East	Online	974719.44	73.18	2516.08322	4496.26	1	high value
355.74	0.1	3260.83	East	Online	896820.54	73.18	2516.08322	3260.83	1	high value
41.97	0.2	862.02	North	Online	105806.37	73.18	2516.08322	862.02	1	low value
299.23	0.23	2075.55	Central	Mobile App	754358.83	73.18	2516.08322	2075.55	1	high value
236.93	0.18	4864.43	North	Online	597300.53	73.18	2516.08322	4864.43	1	high value
55.59	0.12	3502.59	West	Online	140142.39	73.18	2516.08322	3502.59	1	high value
265.88	0.06	4909.36	East	Online	670283.48	73.18	2516.08322	4909.36	1	high value
420.77	0.23	3148.63	South	Online	1060761.17	73.18	2516.08322	3148.63	1	high value
387.55	0.19	2672.73	West	Online	977013.55	73.18	2516.08322	2672.73	1	high value
481.8	0.02	618.83	West	Online	1214617.8	73.18	2516.08322	618.83	1	low value
159.42	0.07	1147.23	East	Mobile App	401897.82	73.18	2516.08322	1147.23	1	low value
374.19	0.2	341.17	Central	Online	943332.99	73.18	2516.08322	341.17	1	low value
105.83	0.14	4194	East	Online	266797.43	73.18	2516.08322	4194	1	high value



8.Create a measure called "Date Difference" that calculates the number of days between the latest Order Date and today's date using the DATEDIFF function.

Answer

```
1 Datediff = DATEDIFF(amazon_store_sales_2022[OrderDate].[Date], TODAY(),DAY)
```

OrderID	OrderDate	ProductID	ProductCategory	Quantity	UnitPrice	Discount	SalesAmount	Region	SalesChannel	totalsales online	Datediff	today
10004	04 January 2022	2072	Beauty	7	260.46	0.22	1697.47	East	Online	1697.47	1637	29 June
10006	06 January 2022	2021	Beauty	1	287.8	0.07	4592.68	West	Mobile App		1635	29 June
10008	08 January 2022	2087	Beauty	2	274.17	0.03	3802.72	North	Online	3802.72	1633	29 June
10011	11 January 2022	2088	Beauty	9	263.02	0.26	3265.31	East	Online	3265.31	1630	29 June
10015	15 January 2022	2022	Beauty	1	84.4	0.02	4401.51	South	Online	4401.51	1626	29 June
10018	18 January 2022	2088	Beauty	8	465.39	0.1	582.46	West	Online	582.46	1623	29 June
10021	21 January 2022	2002	Beauty	3	98.66	0.07	2112.19	South	Mobile App		1620	29 June
10030	30 January 2022	2049	Beauty	8	5.55	0.17	4221.81	Central	Mobile App		1611	29 June
10031	31 January 2022	2091	Beauty	8	280.31	0.12	1284.79	West	Mobile App		1610	29 June
10044	13 February 2022	2064	Beauty	5	14.12	0.05	4334.6	South	Mobile App		1597	29 June
10047	16 February 2022	2007	Beauty	3	199.11	0.17	361.16	West	Mobile App		1594	29 June
10055	24 February 2022	2014	Beauty	8	386.64	0.25	4496.26	East	Online	4496.26	1586	29 June
10056	25 February 2022	2009	Beauty	4	355.74	0.1	3260.83	East	Online	3260.83	1585	29 June
10059	28 February 2022	2002	Beauty	7	41.97	0.2	862.02	North	Online	862.02	1582	29 June
10065	06 March 2022	2008	Beauty	7	299.23	0.23	2075.55	Central	Mobile App		1576	29 June
10074	15 March 2022	2006	Beauty	4	236.93	0.18	4864.43	North	Online	4864.43	1567	29 June
10076	17 March 2022	2004	Beauty	4	55.59	0.12	3502.59	West	Online	3502.59	1565	29 June
10078	19 March 2022	2093	Beauty	8	265.88	0.06	4909.36	East	Online	4909.36	1563	29 June
10081	22 March 2022	2090	Beauty	2	420.77	0.23	3148.63	South	Online	3148.63	1560	29 June
10082	23 March 2022	2044	Beauty	3	387.55	0.19	2672.73	West	Online	2672.73	1559	29 June
10086	27 March 2022	2100	Beauty	5	481.8	0.02	618.83	West	Online	618.83	1555	29 June
10094	04 April 2022	2062	Beauty	2	159.42	0.07	1147.23	East	Mobile App		1547	29 June
10097	07 April 2022	2080	Beauty	8	374.19	0.2	341.17	Central	Online	341.17	1544	29 June
10105	15 April 2022	2029	Beauty	9	105.83	0.14	4194	East	Online	4194	1536	29 June
10111	21 April 2022	2009	Beauty	7	399.41	0.13	3300.95	North	Online	3300.95	1530	29 June
10120	30 April 2022	2035	Beauty	6	127.53	0.06	4780.01	North	Online	4780.01	1521	29 June
10121	01 May 2022	2035	Beauty	7	281.41	0.02	145.21	Central	Mobile App		1520	29 June
10127	07 May 2022	2073	Beauty	9	219.46	0.08	1717.26	West	Mobile App		1514	29 June

amazon_store_sales_2022 (500 rows) Column: Datediff (500 distinct values)